quantdesk: Building and Testing an Options Research Desk from First Principles

Mohammed Alazemi
BSc Physics, The University of Manchester
github.com/Mohammed-Alazemi/quantdesk

October 2026

Abstract

This paper describes `quantdesk`, a Python library I wrote to turn an options trade idea into a decision: what it is worth, what it is exposed to, how much to risk, and whether the evidence for it is real. It covers:

- Black–Scholes pricing and the Greeks;
- volatility forecasting with GARCH and smile fitting with SVI;
- semi-analytic Heston pricing and calibration;
- a delta-hedging simulator;
- risk-neutral densities, variance swaps and earnings event volatility;
- Kelly sizing;
- backtest statistics that correct for luck and for trying many ideas;
- an inventory-aware market maker.

The rule throughout was the one I learned in physics: *a model is not trusted until it reproduces a result whose answer is known.* Each of the 43 automated tests checks the code against a closed-form result, an independent numerical method, or a published table. I explain the mathematics behind each part, how I built and checked it, and what it taught me. The most useful lesson was that the code works, but the earnings strategy I trade does not yet have a proven edge.

1 Why I built it

I trade listed options on US large caps to a written rulebook. My first tool was a single script: Black–Scholes prices, a GARCH volatility forecast, a Monte Carlo of the trade, Kelly sizing and a seven-gate checklist. It gave confident answers, but nothing in it showed whether those answers were right. This project rebuilt that script as a tested library and extended it until each question a trader should ask had a tool behind it:

- What is this option worth?
- What does the market's smile imply?
- Where does the P&L actually come from?
- How much should I risk?
- Is my backtest lying to me?

2 Pricing: Black–Scholes, the Greeks and implied volatility

Model. The stock follows geometric Brownian motion, $dS = \mu S\,dt + \sigma S\,dW$. Itô's lemma applied to an option value $V(S,t)$ gives

$$dV = \Big(\partial_t V + \mu S\,\partial_S V + \tfrac{1}{2}\sigma^2 S^2 \partial_{SS} V\Big)dt + \sigma S\,\partial_S V\,dW. \tag{1}$$

Hold the option and short $\Delta = \partial_S V$ shares. The dW terms cancel, so the portfolio is riskless and must earn the risk-free rate r. That gives the Black–Scholes equation

$$\partial_t V + \tfrac{1}{2}\sigma^2 S^2 \partial_{SS} V + (r-q) S\,\partial_S V - rV = 0. \tag{2}$$

The drift μ has disappeared: an option prices volatility, not direction. For a call with dividend yield q,

$$C = Se^{-qT}N(d_1) - Ke^{-rT}N(d_2), \qquad d_{1,2} = \frac{\ln(S/K) + (r - q \pm \tfrac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}. \tag{3}$$

What I built.

- **Prices and Greeks.** Vectorised prices and seven Greeks: Δ, Γ, vega, Θ, ρ, vanna and volga.
- **Implied volatility.** A solver that first checks the no-arbitrage bounds

  $$\max(Se^{-qT} - Ke^{-rT},\, 0) < C < Se^{-qT},$$

  then finds the root with Brent's method. It returns NaN instead of a meaningless number when a quote violates the bounds.
- **American options.** A Cox–Ross–Rubinstein binomial tree with early exercise.

How I checked it. Table 2 lists every check in the library. The pricing checks are:

- put–call parity holds to 10^{-14};
- the textbook price $C(100, 100, 1, 5\%, 20\%) = 10.4506$ is reproduced;
- implied vol round-trips to machine precision;
- every Greek matches a central finite difference to 10^{-6}.

The American put is worth 10.377, against 10.109 for the European, so the early-exercise premium is positive, as it must be. The American call with no dividends equals the European call, as theory requires.

3 Volatility: GARCH forecasts and the SVI smile

GARCH(1,1). Daily variance follows $h_t = \omega + \alpha r_{t-1}^2 + \beta h_{t-1}$. I fit it by maximum likelihood, $\min \tfrac{1}{2}\sum(\ln h_t + r_t^2/h_t)$, with *variance targeting*: I fix $\omega = V(1-\alpha-\beta)$ at the sample variance V and fit only (α, β). A free three-parameter fit on a few hundred returns often wandered to absurd values; the two-parameter fit is stable.

The forecast of average variance over a horizon of H days uses the mean reversion $h_{t+k} = V + (\alpha+\beta)^k(h_{t+1} - V)$. This forecast, not last month's realised vol, is what an option's implied vol should be compared against. To check the fit, I simulated a GARCH process with known parameters: the fit recovers α to within 0.02 and β to within 0.03.

SVI. Gatheral's raw SVI parametrises total implied variance $w = \sigma_{\text{imp}}^2 T$ in log-moneyness $k = \ln(K/F)$:

$$w(k) = a + b\big(\rho(k-m) + \sqrt{(k-m)^2 + s^2}\big). \tag{4}$$

A fitted smile can still contain *butterfly arbitrage*, meaning a negative probability density. I implemented Gatheral's condition $g(k) \geq 0$, which flags such smiles.

4 Stochastic volatility: semi-analytic Heston

Black–Scholes assumes constant volatility, but real smiles are skewed: downside strikes trade at higher implied vol. Heston's model lets variance v follow its own mean-reverting process, correlated with the stock:

$$dS = (r-q)S\,dt + \sqrt{v}\,S\,dW_1, \qquad dv = \kappa(\theta - v)\,dt + \xi\sqrt{v}\,dW_2, \qquad dW_1 dW_2 = \rho\,dt. \tag{5}$$

Pricing. The characteristic function $\varphi(u) = \mathbb{E}^{\mathbb{Q}}[e^{iuX}]$ of $X = \ln(S_T/F)$ is known in closed form. I used the "little trap" version of Albrecher et al., which avoids a branch-cut discontinuity in Heston's original expression at long maturities:

$$\varphi(u) = \exp\Big\{\tfrac{\kappa\theta}{\xi^2}\Big[(a-d)T - 2\ln\tfrac{1-ge^{-dT}}{1-g}\Big] + \tfrac{v_0}{\xi^2}(a-d)\tfrac{1-e^{-dT}}{1-ge^{-dT}}\Big\}, \tag{6}$$
$$a = \kappa - \rho\xi iu, \quad d = \sqrt{a^2 + \xi^2(iu+u^2)}, \quad g = \tfrac{a-d}{a+d}.$$

Prices then follow from Lewis's single integral,

$$C = Se^{-qT} - \frac{\sqrt{FK}\,e^{-rT}}{\pi}\int_0^\infty \frac{\operatorname{Re}\big[e^{iu\ln(F/K)}\varphi(u-\frac{i}{2})\big]}{u^2+\frac{1}{4}}\,du. \tag{7}$$

I evaluate the integral on 256 fixed Gauss–Legendre nodes after the substitution $u = c\,t/(1-t)$. The scale c is set by the total variance, so the nodes sit where the integrand lives for both short and long maturities. Fixed nodes make the price a smooth function of the parameters, which matters for calibration. Pricing 30 strikes takes about 0.3 ms.

Calibration. I fit $(v_0, \kappa, \theta, \xi, \rho)$ by least squares on price errors divided by Black–Scholes vega. To first order this is an implied-vol error, but it avoids inverting every model price inside the optimiser. The fit runs from several starting points to avoid local minima.

How I checked it. I tested the pricer four ways:

- it equals Black–Scholes to 10^{-6} in the limit $\xi \to 0$, $\rho = 0$;
- it matches adaptive quadrature to 10^{-8};
- it agrees with my independent Monte Carlo simulator within its standard error;
- it satisfies put–call parity.

Calibrating to a surface the model generated itself recovers all five parameters to a relative error of 10^{-4}. Figure 1 shows the harder test: 45 quotes over five expiries with 0.2 vol points of noise added.

5 Where an option trader's P&L comes from

This section was the most valuable to me, because it explains my own trades. Buy an option at implied vol σ_i and delta-hedge it while the stock moves with realised vol σ_r. Over dt, the hedged position (option minus Δ_i shares, financed at r) gains

$$d\Pi = \tfrac{1}{2}\Gamma_i S^2\big(\sigma_r^2 - \sigma_i^2\big)\,dt, \tag{8}$$

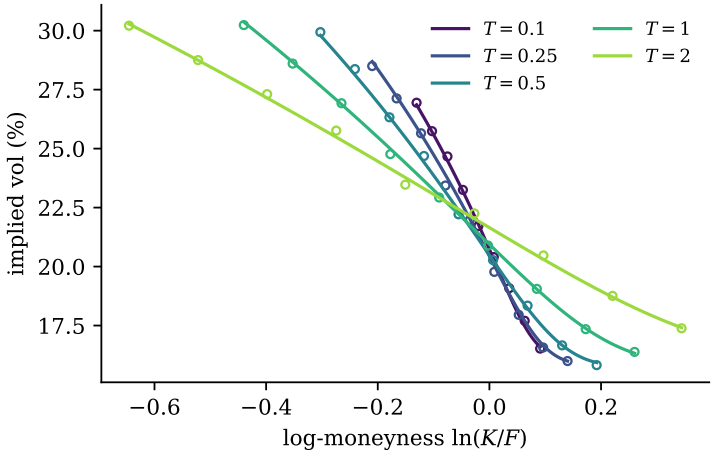


Figure 1: Heston calibrated to 45 noisy implied-vol quotes (circles) across five expiries (lines are the fitted model). The fitted parameters $(v_0, \kappa, \theta, \xi, \rho) = (0.045, 2.55, 0.060, 0.69, -0.71)$ compare with the true $(0.045, 2.50, 0.060, 0.70, -0.70)$. The residual is 0.16 vol points, just below the 0.2-point noise level, which is what a correct fit should give.

because the option's theta exactly pays for gamma at the implied vol. So the hedged P&L is a bet on realised against implied volatility. There are two hedging choices (Wilmott, ch. 10):

- **Hedge at σ_r.** The P&L is deterministic in the continuous limit: $[V(\sigma_r) - V(\sigma_i)]\, e^{rT}$. The edge is locked in.
- **Hedge at σ_i.** The expected P&L has the same sign, but it is $\int e^{r(T-t)} \frac{1}{2} \Gamma_i S^2 (\sigma_r^2 - \sigma_i^2) dt$. That depends on the path, because Γ is large only while the stock stays near the strike.

Real hedging is discrete. With n rebalances and no vol edge, each step leaves an error proportional to $\Gamma S^2 \sigma^2 (Z^2 - 1) \Delta t$. Kamal and Derman showed that the total has standard deviation $\approx \sqrt{\pi/4}\, \nu\, \sigma / \sqrt{n}$ for an at-the-money option, where ν is vega.

What I built and found. I wrote a vectorised hedging simulator (20,000 antithetic paths, optional transaction costs). In it, a 6-month ATM call is bought at $\sigma_i = 20\%$ while the stock realises $\sigma_r = 30\%$. Figure 2 shows four results:

- **Hedging at σ_r converges to the locked-in \$2.82.** The spread halves each time the number of rebalances is multiplied by four.
- **Hedging at σ_i has the same mean, but its spread levels off near \$1.3** however often you rebalance: that remainder is the path dependence.
- **With no edge, the simulated hedging error matches Derman–Kamal** to within a few percent at every n.
- **The hedged P&L does not depend on the stock's drift:** the means at $\mu = -30\%$, 0% and $+40\%$ agree to within 3 cents.

6 What the smile prices in

Risk-neutral density. Differentiating $C(K) = e^{-rT} \int (S - K)^+ f(S)\, dS$ twice with respect to the strike gives Breeden and Litzenberger's result

$$f(K) = e^{rT} \frac{\partial^2 C}{\partial K^2}. \tag{9}$$

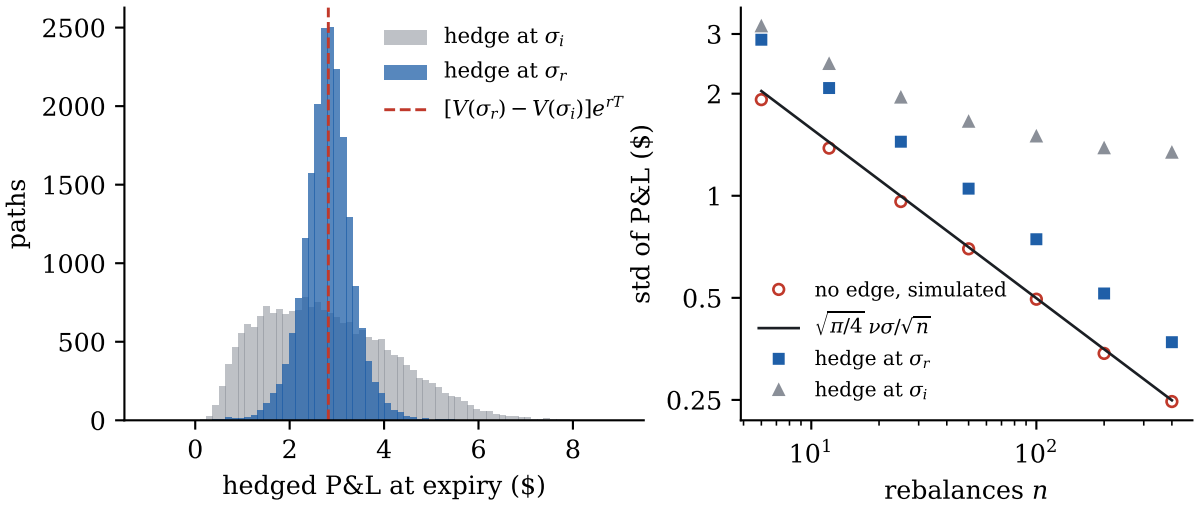


Figure 2: Delta hedging an ATM call bought at 20% implied vol while the stock realises 30%. **Left:** P&L after 200 rebalances. Hedging at the realised vol concentrates it on the locked-in value (dashed); hedging at the implied vol leaves a wide, path-dependent spread. **Right:** the standard deviation of P&L against the number of rebalances. Circles: no vol edge, compared with the Derman–Kamal formula (line).

The option market therefore prices a full probability distribution, not only a volatility. I take the second difference of prices generated from a smooth SVI smile. With flat volatility, it recovers the lognormal density to 10^{-6}. For the skewed smile in Fig. 3, the density integrates to 1.0000 and has the forward as its mean, with skewness -0.68 and excess kurtosis 3.1. That is a fat left tail which a single volatility number cannot describe.

Variance swap. The payoff of a log contract can be replicated with a static strip of options. Since $\int \sigma^2 dt = 2\big(\int dS/S - \ln(S_T/S_0)\big)$, the fair strike of a variance swap is

$$K_{\text{var}} = \frac{2e^{rT}}{T}\Big[\int_0^F \frac{P(K)}{K^2}dK + \int_F^\infty \frac{C(K)}{K^2}dK\Big]. \tag{10}$$

Checks: flat vol returns σ^2 to 10^{-4}. On the skewed smile the variance-swap volatility is 28.3%, against 22.4% at the money. The difference is what the skew costs.

Earnings event volatility. Variance adds across days. If an expiry n trading days away contains one event day, then $\sigma_{\text{term}}^2\, n = \sigma_E^2 + \sigma_{\text{reg}}^2(n-1)$. With a term IV of 36%, a normal-day vol of 24% and $n = 40$, the event day carries $\sigma_E = 171\%$ annualised. That implies an expected absolute move of $\sigma_E\sqrt{2/\pi}/\sqrt{252} = 8.6\%$. Removing σ_E after the print predicts the IV crush before it happens. This is the arithmetic behind my rule "never hold through earnings".

7 From a model to a decision: sizing

Monte Carlo under the real-world measure. Prices are expectations under the risk-neutral measure $\mathbb{Q}$, where every asset drifts at r. My P&L, however, happens under the real-world measure $\mathbb{P}$, with drift μ. The library therefore prices the option with market implied vol and simulates the trade under my own drift and volatility forecast. The simulation uses GBM, or Heston with a path-dependent exit IV, and every fill pays half the bid–ask spread plus commission.

Kelly. For a return on premium R, the growth-optimal fraction maximises $g(f) = \mathbb{E}[\ln(1+fR)]$. For a binary bet that wins b with probability p, the optimum is $f^* = p - (1-p)/b$. Option returns

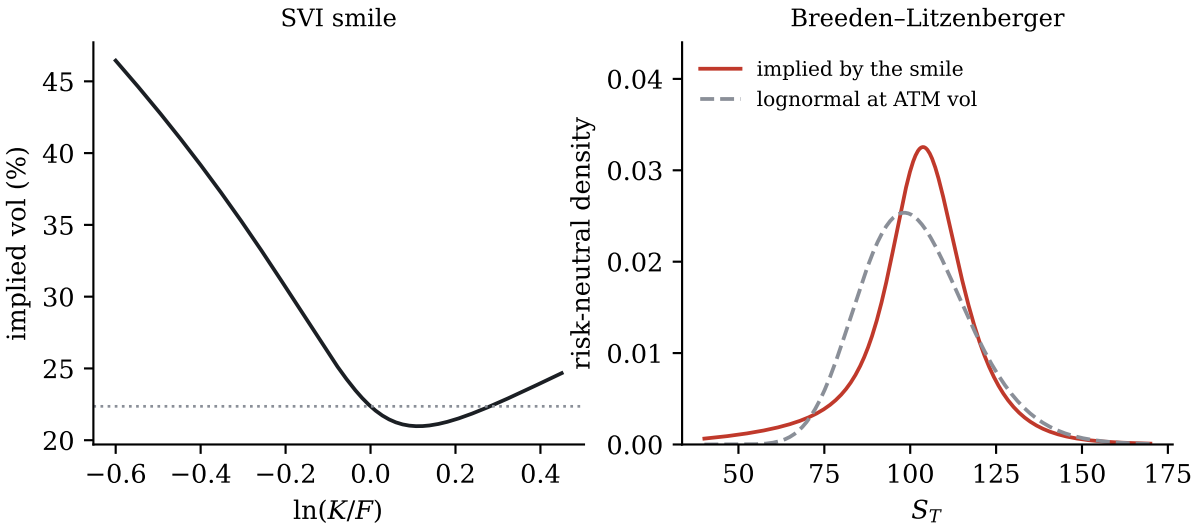


Figure 3: Left: a skewed SVI smile (dotted line: ATM vol). **Right:** the risk-neutral density it implies (Breeden–Litzenberger), against a lognormal with the same ATM vol. The skew moves probability into the left tail and makes the centre narrower.

are skewed and bounded at -100%, so the library maximises g directly over the simulated returns: a grid search refined by a bounded one-dimensional search. On a bet that wins $1.5\times$ or loses $1\times$ at $p = \frac{1}{2}$, it returns 0.1655 against the exact $\frac{1}{6}$; the difference is sampling noise.

Since $g(f) \approx f\mu - \frac{1}{2}f^2\sigma^2$, half-Kelly keeps 75% of the growth with half the volatility, and that is the default. A bootstrap of f^* shows how fragile the estimate is. A sweep over assumed drifts shows whether an apparent edge exists only under an optimistic drift. In my demo it did: the Kelly fraction is 0% at every drift up to 20% a year.

The rulebook. My trading rules are an editable `Rules` object with eight hard gates:

1. probability of profit;
2. Sharpe per trade;
3. Kelly fraction above zero;
4. IV no more than 15% above HV30;
5. at least 45 days to expiry;
6. exit before earnings;
7. position no larger than 10% of the portfolio;
8. a limit on the number of open positions.

Soft gates warn. A size is suggested only if every hard gate passes.

8 Is the backtest lying?

The strategy and its result. My main strategy buys an at-the-money call about six weeks before earnings and sells it the day before the print, to catch the run-up in implied vol. I backtested it on synthetic markets in which I control the size of the pre-earnings IV "pump". The backtest uses realistic costs and 20 tickers × 39 events. Table 1 shows the result:

- **With a 35% pump, the strategy only breaks even.** Theta and costs eat the gain.
- **With no pump, it loses** 9.2% **per trade**, a loss that is clearly significant.

Synthetic market	trades	mean return on premium (95% CI)	t	$P(\text{Sharpe} > 0)$
No IV pump	780	−9.2% (−15.1%, −3.2%)	−3.04	0.3%
35% IV pump	780	−0.2% (−5.8%, +5.2%)	−0.08	47%
Expiry after the print (35% pump): 14 d −8.9% — 45 d −0.2% — 90 d +4.7% — 120 d +6.3%				

Table 1: Pre-earnings IV-pump backtest with costs, on synthetic data where the pump is put in by construction.

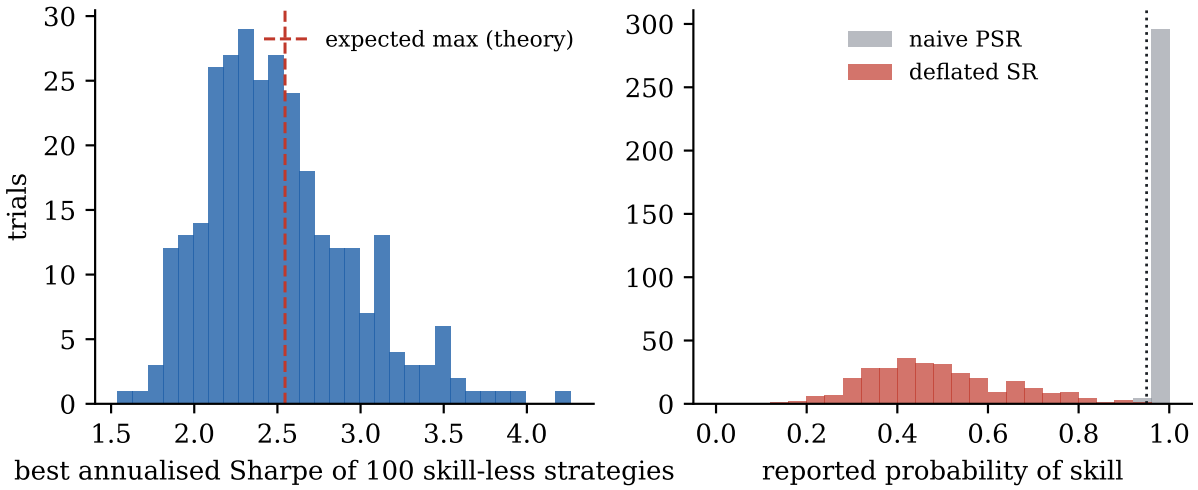


Figure 4: Selection bias. **Left:** the best annualised Sharpe among 100 skill-less strategies (250 days each), over 300 trials, with the theoretical expected maximum marked. **Right:** the probability of skill reported for that best strategy. The naive PSR almost always passes the 95% line (dotted); the deflated Sharpe ratio never does.

- **A null test that places the same trade on random dates** shows whether timing adds anything at all.

Overfitting. Longer expiries look better in Table 1, but I found that by trying several. Trying many variants and keeping the best inflates the Sharpe ratio. Two statistics from Bailey and López de Prado correct for this. The *probabilistic Sharpe ratio* is

$$\mathrm{PSR}(SR^*) = N\left(\frac{(\widehat{SR} - SR^*)\sqrt{n-1}}{\sqrt{1 - \gamma_3 \widehat{SR} + \frac{\gamma_4 - 1}{4}\widehat{SR}^2}}\right), \tag{11}$$

which accounts for sample length, skewness γ_3 and kurtosis γ_4. The *deflated* Sharpe ratio sets the benchmark SR^* to the expected maximum Sharpe of N skill-less trials,

$$SR^* = \sigma_{SR}\Big[(1-\gamma)N^{-1}(1-\tfrac{1}{N}) + \gamma N^{-1}(1-\tfrac{1}{Ne})\Big], \tag{12}$$

where γ is the Euler–Mascheroni constant. Figure 4 shows the test I used. Among 100 random strategies, the best has a median annualised Sharpe of 2.4. The naive test calls it skilled in 99.7% of trials; the deflated test does so in none of 300 trials. The library also provides walk-forward splits with an embargo, so that parameters are never tuned on the period they are tested on.

9 Market making with inventory risk

A market maker earns the spread but carries inventory q. Avellaneda and Stoikov model fills as arriving at rate $Ae^{-k\delta}$ for a quote δ away from the mid, and give the market maker exponential

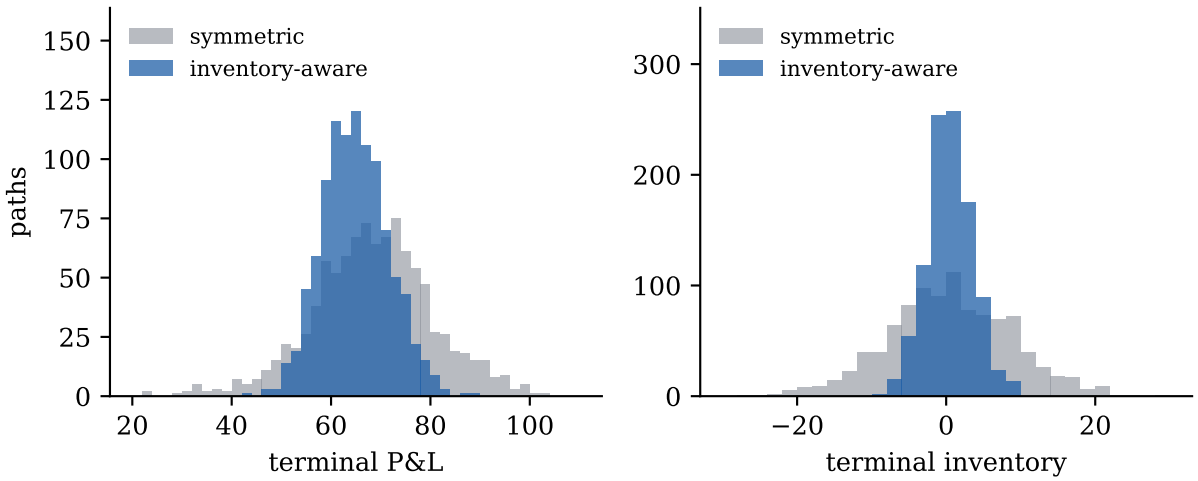


Figure 5: Avellaneda–Stoikov market making ($\gamma = 0.1$, 1000 simulated sessions). Skewing quotes against inventory keeps inventory near zero (right) and makes the P&L distribution far tighter (left).

utility with risk aversion γ. The optimal quotes centre on a reservation price that shifts against inventory:

$$r = s - q\gamma\sigma^2(T-t), \qquad \delta_a + \delta_b = \gamma\sigma^2(T-t) + \tfrac{2}{\gamma}\ln\!\big(1+\tfrac{\gamma}{k}\big). \tag{13}$$

With their parameters, my simulator reproduces the paper's comparison (Fig. 5). Both strategies quote a spread of 1.49:

- **inventory-aware quoting:** P&L 64.9 ± 6.6, final inventory spread 3.0;
- **symmetric quoting around the mid:** P&L 68.5 ± 13.2, final inventory spread 8.4.

Managing inventory gives up about 5% of the mean profit and halves the risk.

10 How I built it

- **A known answer before any trust.** Every module was paired with a test against an exact result, an independent method (quadrature, Monte Carlo, finite differences) or a published table (Table 2). Tests run on every change through GitHub Actions, on two Python versions.
- **Variance reduction and reproducibility.** Antithetic variates in every simulation, fixed random seeds, and checks that stochastic tests pass across several seeds, so that no test passes by luck.
- **Vectorised NumPy throughout.** The full test suite runs in about 11 seconds.
- **Honest defaults.** Costs are on by default; the IV solver returns NaN when a quote violates the arbitrage bounds; no size is suggested unless every hard gate passes.

11 What I learned

1. **An option is a bet on volatility with a direction attached.** The hedging experiment made $\frac{1}{2}\Gamma S^2(\sigma_r^2 - \sigma_i^2)$ concrete: I can be right about direction and still lose to theta and IV crush.
2. **Code that runs is not code that is right.** Most of my early errors were not in the formulas but in details. One example: a test of my Kelly optimiser kept failing until I realised that the grid itself limited its accuracy to 0.3%.

Component	Check	Result
Black–Scholes	put–call parity; textbook value	1.4×10^{-14}; 10.4506
Greeks (7)	central finite differences	$< 10^{-6}$
Implied vol	round trip; arbitrage bounds	2×10^{-16}; NaN outside bounds
American CRR	early-exercise premium; call = European	$10.377 > 10.109$; within 5×10^{-3}
GARCH(1,1)	recover simulated (α, β)	within $(0.02, 0.03)$
SVI	recover a curve; detect arbitrage	RMSE $< 10^{-5}$; flags spiky smile
Heston	BS limit; quadrature; Monte Carlo	10^{-6}; 10^{-8}; within 4 s.e.
Heston calibration	noise-free / noisy surface	params to 10^{-4} / RMSE 0.16 vol pts
Delta hedging	locked-in P&L; Derman–Kamal; drift	\$2.82; within a few %; within 3 cents
Densities	BL vs lognormal; var swap at flat vol	10^{-6}; 10^{-4} relative
Event vol	desk worked example	171%, move 8.6%
Kelly	analytic $f^* = 1/6$	0.1655
Deflated Sharpe	false positives, best of 100	naive $> 90\%$, deflated $\leq 5\%$
Market making	Avellaneda–Stoikov (2008) table	spread 1.49; risk halved

Table 2: What each test checks. All 43 pass.

3. **The market prices a distribution, not a number.** The skew is information: it costs money in a variance swap and shows a fat left tail in the density.
4. **The backtest is the most dangerous part.** Costs alone can turn a real pump into a loss, and searching over variants creates a Sharpe ratio from nothing. On synthetic data where I put the edge in myself, my strategy only broke even. That is the most important result in this paper, and it changes how I trade.
5. **Physics habits transfer directly:** validating against exact solutions, checking convergence, and suspecting the numerics before the physics.

12 Limitations and next steps

- **The backtest uses synthetic data.** The next step is real historical option implied volatility, run with walk-forward splits and judged by the deflated Sharpe ratio.
- **The Monte Carlo exit IV is an at-the-money proxy,** not a full repricing of the surface.
- **Heston is calibrated to synthetic surfaces only.** Fitting it to a live chain, and comparing it with SVI, is the natural extension.
- **Smaller limits.** The scenario P&L handles debit positions only, and the backtest uses one constant-maturity IV series.

This is research code for learning, not investment advice.

References

[1] F. Black and M. Scholes, "The pricing of options and corporate liabilities," *J. Political Economy* 81 (1973).

[2] S. Heston, "A closed-form solution for options with stochastic volatility," *Rev. Financial Studies* 6 (1993).

[3] H. Albrecher, P. Mayer, W. Schoutens and J. Tistaert, "The little Heston trap," *Wilmott Magazine* (2007).

[4] A. Lewis, *Option Valuation under Stochastic Volatility* (Finance Press, 2000).

[5] J. Gatheral, *The Volatility Surface* (Wiley, 2006).

[6] D. Breeden and R. Litzenberger, "Prices of state-contingent claims implicit in option prices," *J. Business* 51 (1978).

[7] P. Carr and D. Madan, "Towards a theory of volatility trading," in *Volatility* (Risk Books, 1998).

[8] S. Kamal and E. Derman, "When you cannot hedge continuously," *Risk* (1999).

[9] P. Wilmott, *Paul Wilmott Introduces Quantitative Finance*, 2nd ed. (Wiley, 2007).

[10] D. Bailey and M. López de Prado, "The Sharpe ratio efficient frontier," *J. Risk* 15 (2012); "The deflated Sharpe ratio," *J. Portfolio Management* 40 (2014).

[11] M. Avellaneda and S. Stoikov, "High-frequency trading in a limit order book," *Quantitative Finance* 8 (2008).

[12] T. Bollerslev, "Generalized autoregressive conditional heteroskedasticity," *J. Econometrics* 31 (1986).